

2016.0 RANGE ROVER (LG), 211-02

# POWER STEERING

DESCRIPTION AND OPERATION

---

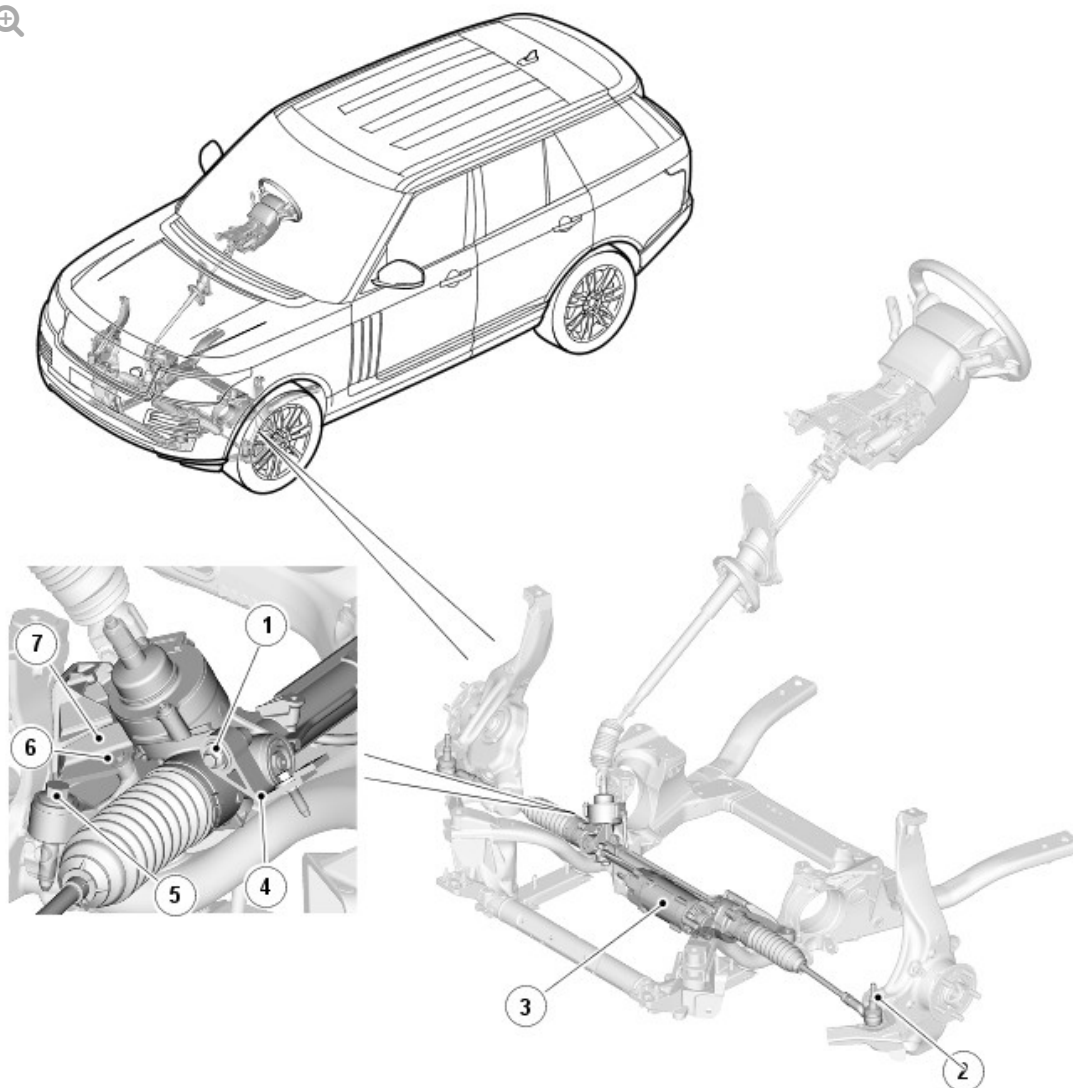
---

## COMPONENT LOCATION

### Component Location - 1 of 1

#### NOTE:

RHD (right-hand drive) installation shown, LHD (left-hand drive) installation similar.



E142565

| ITEM |  | DESCRIPTION                 |
|------|--|-----------------------------|
| 1    |  | M8 bolt and washer (2 off)  |
| 2    |  | M14 nut and washer (2 off)  |
| 3    |  | Steering gear               |
| 4    |  | NVH bracket                 |
| 5    |  | M14 bolt and washer (2 off) |
| 6    |  | M6 bolt and washer          |
| 7    |  | Mass damper                 |

## OVERVIEW

The vehicle is fitted with Electric Power Steering (EPS), provided by a steering gear unit with variable ratio, rack and pinion steering and speed sensitive power assistance. Power assistance is from an electric motor attached to the steering gear.

The steering gear also incorporates:

- A power steering control module, on the end of the electric motor.
- A reduction gear.
- A torque sensor.

Power assistance is controlled by the power steering control module, which uses the electric motor to apply an axial force on the steering rack via a reduction gear. The level of power assistance depends on the steering torque applied with the steering wheel, the vehicle speed and the terrain response mode selected. The power steering control module also uses the electric motor to provide the following assistance functions:

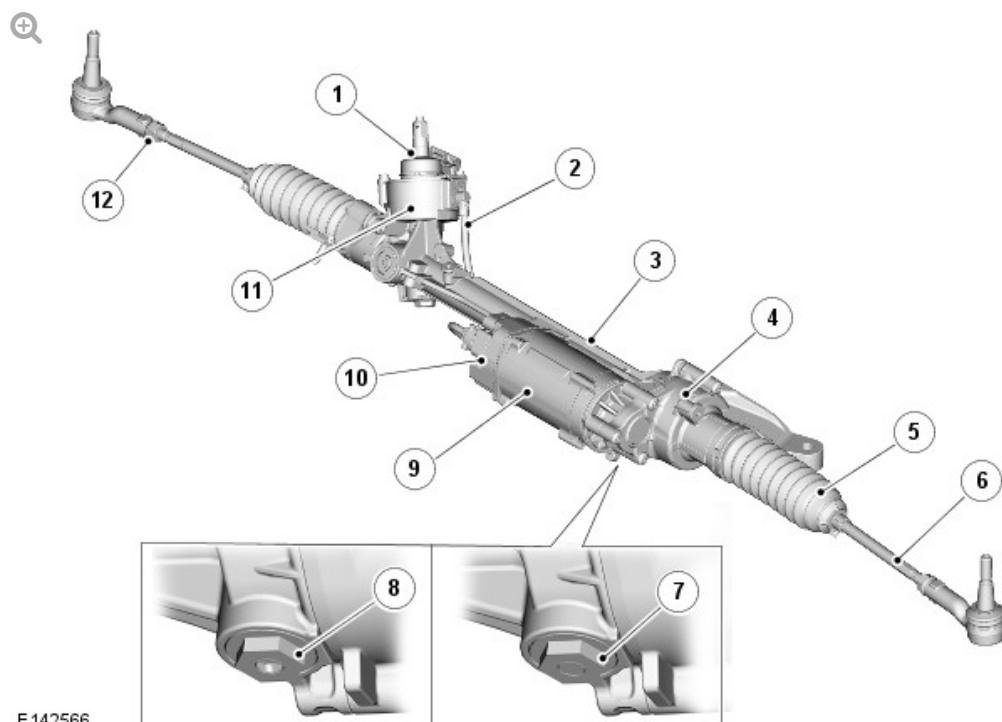
- Assistance changes during Eco Stop/Start functions (If fitted).
- Assistance reduction if system overheats.
- Assistance reduction if low or high voltage.
- Park assist support (If fitted).

For additional information, refer to: [Parking Aid](#) (413-13 Parking Aid, Description and Operation).

---

## DESCRIPTION

### STEERING GEAR



| ITEM | DESCRIPTION                             |
|------|-----------------------------------------|
| 1    | Input shaft                             |
| 2    | Signal and power lead for torque sensor |
| 3    | Steering gear housing                   |
| 4    | Reduction gear                          |
| 5    | Boot (2 off)                            |
| 6    | Tie rod (2 off)                         |
| 7    | Drain valve (normal condition) - closed |
| 8    | Drain valve (deployed) - open           |
| 9    | Electric motor                          |
| 10   | Power steering control module           |
| 11   | Torque sensor                           |

The steering gear is installed on the front subframe. Two bosses on the rear of the steering gear are attached direct to the subframe with M14 bolts and captive washers. Two bosses on top of the steering gear are attached to NVH (noise, vibration and harshness) brackets on the subframe with M8

bolts and washers. A mass damper is attached to the steering gear with an M6 bolt and washer.

The input shaft of the steering gear connects to the pinion extension of the steering column lower shaft.

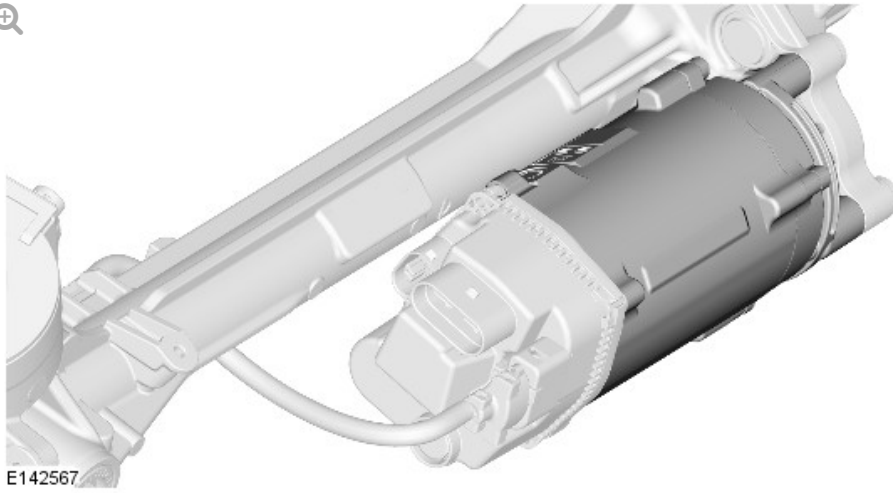
For additional information, refer to: [Steering Column](#) (211-04 Steering Column, Description and Operation).

The steering rack is installed in a steering gear housing that also contains the pinion shaft and the torque sensor. A torsion bar connects the pinion shaft to the input shaft. Ball jointed tie rods connect the ends of the steering rack to the knuckles of the front suspension. The ends of the steering rack are protected by corrugated boots.

The steering gear housing incorporates a mounting for the electric motor. A reduction gear housing, on the end of the steering gear housing, contains a recirculating ball nut installed on a threaded section of the steering rack. A toothed belt transmits drive from the electric motor to the recirculating ball nut to provide the power assistance. The ratio of electric motor revolutions to steering wheel revolutions is approximately 20:1.

A water drain valve, consisting of a spring and plunger, is installed in the mounting for the electric motor. The drain valve is held closed by a salt tablet. If the salt tablet is exposed to water from the inside of the rack, it dissolves, which allows the spring to move the plunger to the open position to drain the water. A visual inspection of the water drain valve should be made at regular intervals. If the valve is open the steering gear must be replaced.

## **Electric Motor**

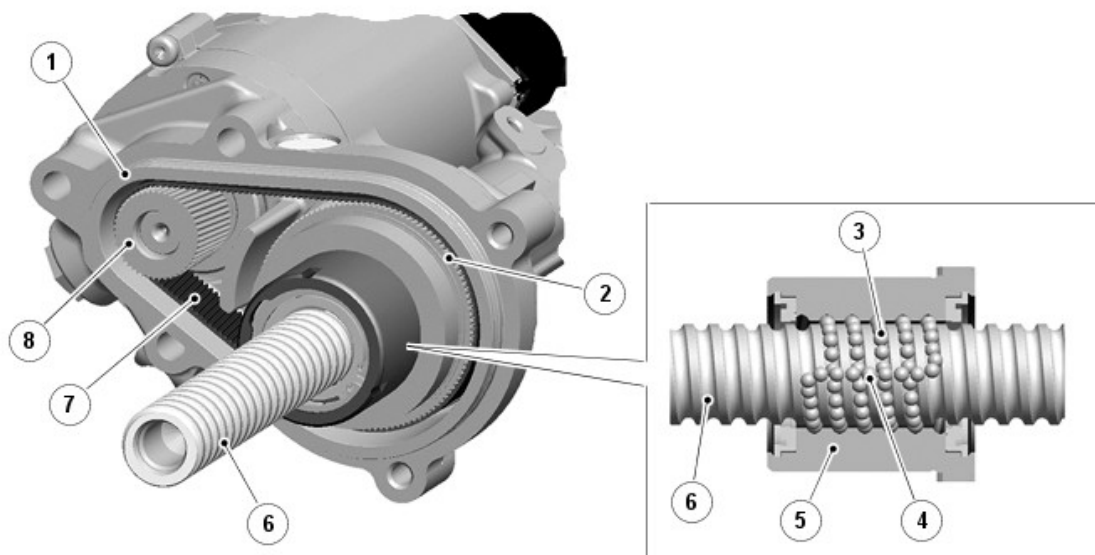


The electric motor is a brushless DC (direct current) motor. The electronics that convert the DC power supply into phase voltages are contained in the power steering control module.

Motor position is detected by two Hall effect sensors in the power steering control module, adjacent to rotating magnets on the electric motor shaft. The two sensors have different resolutions, which allows the power steering control module to monitor both fast and slow movement effectively. From the position of the electric motor, the power steering control module can deduce the position of the road wheels and the steering angle.

After first calibrating the straight-ahead position using the steering angle sensor, the power steering control module subsequently uses the motor position sensor inputs for control of the power steering.

## **Reduction Gear**

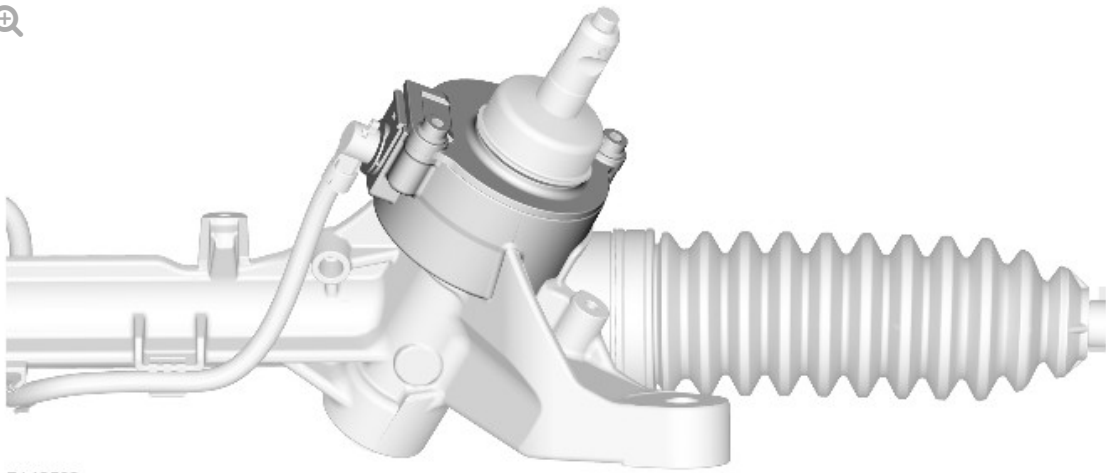


E133530

| ITEM | DESCRIPTION                |
|------|----------------------------|
| 1    | Reduction gear housing     |
| 2    | Ball nut drive pulley      |
| 3    | Ball train                 |
| 4    | Ball return channel        |
| 5    | Ball nut                   |
| 6    | Steering rack ball channel |
| 7    | Drive belt                 |
| 8    | Motor drive pulley         |

The recirculating ball nut in the reduction gear housing is held in a bearing race, allowing the nut to rotate but preventing it from moving axially. As the electric motor rotates the nut, the balls are driven along the channels in the nut and the steering rack, and produce an axial force on the steering rack. A ball return channel in the nut, and mechanisms at either end of the bearing race, recirculate the balls from one side of the nut to the other.

## Torque Sensor

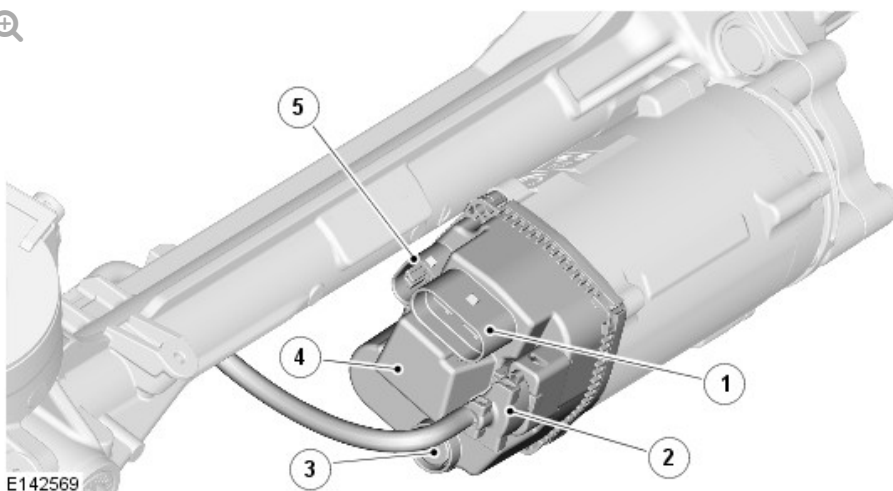


E142568

The torque sensor provides the power steering control module with inputs of the steering torque applied by the driver. The torque sensor is installed at the junction between the input shaft and the pinion shaft, inside the steering gear housing.

Torque applied to the input shaft is transferred to the pinion shaft through the torsion bar, which twists under load. The torque sensor calculates the amount of twist in the torsion bar using two Hall effect sensors to monitor the angular position of the input shaft in relation to the pinion shaft, then transmits the result in a digital signal to the power steering control module.

## Power Steering Control Module



E142569

ITEM

DESCRIPTION



|   |                                                |
|---|------------------------------------------------|
| 1 | Electric motor power supply connector          |
| 2 | Torque sensor connector                        |
| 3 | Diaphragm                                      |
| 4 | Power steering control module cover            |
| 5 | Power steering control module system connector |

The power steering control module is located on the end of the electric motor, under a hermetically sealed cover. A diaphragm in the cover, made from a waterproof breathable fabric, maintains the interior of the cover at ambient pressure, avoiding the need for an open vent.

As well as the control circuitry, the power steering control module also contains the power electronics for operating the electric motor.

Power assistance is provided when the following are present:

- An ignition feed from the ignition relay in the EJB (engine junction box).
- The power mode CAN (controller area network) signal is set to engine running.

Electrical power for operation of the electric motor is provided by a permanent feed from the AJB (auxiliary junction box). For 100% nominal power assistance, supply voltage must be between 10.5 and 18 V. If the supply voltage is outside of these limits, power assistance is either reduced or discontinued.

The power steering control module receives hardwired digital inputs from the torque sensor and high speed CAN bus inputs of:

- Steering angle, from the steering angle sensor.
- Engine speed, from the ECM (engine control module).
- Vehicle speed, from the Anti-lock Braking System (ABS) control module.
- Automatic steering requests, from the parking aid module (if park assist system fitted).

- Terrain mode request from the Terrain Response switch pack.
- Power mode from the CJB (central junction box).
- Eco stop/start system stop request from the CJB (if stop/start system fitted).

The power steering control module uses the input data and software algorithms to control electric motor operation. If required, the power steering control module can also transmit outputs, on the high speed CAN chassis systems, of:

- Warning status, to the instrument cluster.
- Steering status, to the parking aid module (if park assist system fitted) and Terrain Response switch pack.
- Eco stop/start system inhibit requests to the ECM (if stop/start system fitted).
- Tie rod load forces.

#### REDUCED ASSISTANCE WARNING INDICATOR

If there is a significant reduction in power assistance, the amber general warning indicator illuminates and a warning message is displayed in the message center. When the warning message is first displayed, the instrument cluster produces a single alert chime. For additional information, refer to: (413-01 Instrument Cluster)

[Instrument Cluster](#) (Description and Operation),

[Instrument Cluster - SDV6 3.0L Diesel - Hybrid Electric Vehicle](#) (Description and Operation).

If the steering gear is replaced, the correct vehicle specific calibration must be loaded using Land Rover approved diagnostic equipment.

---

## OPERATION

For the EPS (electric power steering) system to provide assistance the

engine must be running.

If a fault occurs which affects control of the electric motor, the power steering control module disables the electric motor. The driver no longer benefits from power assisted steering, but incorrect control of the electric motor is prevented and the vehicle remains fully steerable, although with greater physical effort.

#### VARIABLE POWER ASSISTANCE

At low speeds, when tire resistance to steering inputs is highest, more assistance is applied to reduce the steering wheel torque to comfortable levels. At higher speeds, when tire resistance to steering inputs is less, less assistance is applied.

#### PULL DRIFT COMPENSATION

This feature aids the driver by compensating for the accumulated effects of various factors that contribute to the pull/drift of the vehicle. The feature reduces the steering wheel angle offset that the driver feels while driving in a straight line, by applying a counter torque.

The feature determines the amount of compensation with data from these sensors:

- Steering gear torque sensor
- Steering angle sensor (integrated into the clockspring)
- Yaw rate sensor in the Restraint Control Module (RCM)
- Vehicle speed from the Anti-lock Brake System (ABS) control module.

The feature compensates for pull/drift over time. This feature does not compensate for the instantaneous changes in the road condition or camber.

The amount of compensation will depend on the road conditions. Typically on a 2% camber road this results in approximately 1.75° of steering wheel offset from the straight ahead position.

The feature does not operate, in these circumstances:

- Vehicle speed is low.
- High torque input by the driver through the steering wheel.
- High rate for steering wheel angle change. Large steering input by the driver.
- Vehicle yaw rate exceeds feature limits. For example when cornering.

#### ASSISTANCE CHANGES DURING ECO STOP/START FUNCTIONS (IT FITTED)

This feature actively controls the steering torque feedback to the driver during the engine Eco stopping and starting activities. It integrates the main steering assistance feature with starter motor events, ensuring smooth, progressive, yet timely decreases and increases in power assistance. It is also designed to remove unwanted steering wheel movement during these events, which are not directly instigated by the driver.

#### ASSISTANCE REDUCTION IF SYSTEM OVERHEATS OR VOLTAGE IS TOO HIGH OR LOW

This feature ensures that the steering requirements of the driver continue to be supported during various extreme conditions including:

- Extreme voltage fluctuations (over and under voltage).
- Extreme high temperature.

A temperature sensor is installed in the power steering control module for overload protection of the electric motor. The power steering control module reduces the level of power assistance if the temperature of the electric motor is too high, to reduce the amount of heat generated.

Temperature overload can occur at high ambient temperatures combined with high levels of steering activity, especially when stationary. A temperature overload can also occur if an attempt is made to repeatedly turn the front wheels against a solid object such as a curb.

Power assistance begins to reduce at a power steering control module temperature of approximately 110 °C (230 °F) and reduces to zero at a

temperature of 125 °C (257 °F). At a given level of reduction in the power assistance, the power steering control module records a DTC (diagnostic trouble code) and signals the instrument cluster to produce a warning that assistance has been reduced. When the power steering control module has cooled, normal power assistance is restored and the related instrument cluster warnings are extinguished.

## SOFT END STOPS

The soft end stop feature reduces the level of power assistance shortly before the mechanical end stops are reached. Although the driver will perceive this as increased steering resistance, it makes turning the wheels into full lock much smoother. In addition, this feature reduces the stresses on the mechanical and electrical components of the steering system.

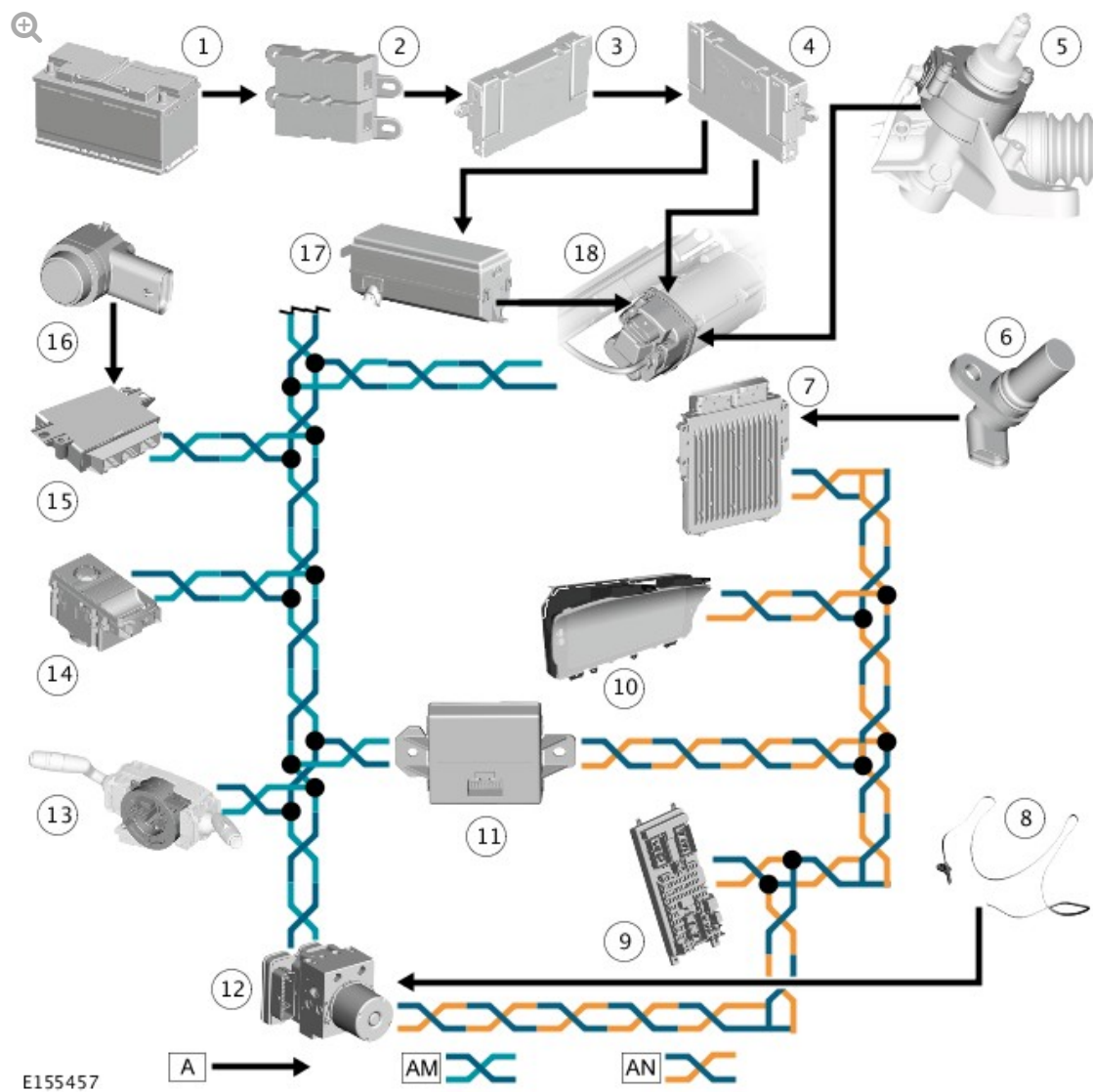
## PARK ASSIST SYSTEM (IF FITTED)

When the park assist system is active, steering trajectory requests from the park distance control module are processed by the power steering control module and applied to the electric motor. This feature assists the driver, during parallel parking maneuvers, by selectively applying turning forces to the steering system to automatically steer the vehicle.

---

## CONTROL DIAGRAM

### Control Diagram - 1 of 1



A = HARDWIRED: AN = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) POWERTRAIN SYSTEMS BUS: AM = HS CAN CHASSIS SYSTEMS BUS.

| ITEM | DESCRIPTION                |
|------|----------------------------|
| 1    | Battery                    |
| 2    | Battery junction box 2     |
| 3    | Battery junction box       |
| 4    | Auxiliary junction box     |
| 5    | Torque sensor              |
| 6    | Crankshaft position sensor |
| 7    | Engine control module      |
|      |                            |

|    |                                                             |
|----|-------------------------------------------------------------|
| 8  | Wheel speed sensor (4 off)                                  |
| 9  | Central junction box                                        |
| 10 | Instrument cluster                                          |
| 11 | Gateway module                                              |
| 12 | Anti-lock Brake System (ABS) control module                 |
| 13 | Steering angle sensor (integrated into clockspring)         |
| 14 | Terrain Response switch pack                                |
| 15 | Park distance control module (if park assist system fitted) |
| 16 | Park assist sensor (4 off)                                  |
| 17 | Engine junction box                                         |
| 18 | Power steering control module                               |